Vedant Shinde

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linkedin.com/in/vedant-nilesh-shinde github.com/Vedant321 Portfolio

Experience

Morgan Stanley *Oct 2021 – Jul 2025*
Senior Associate – Python Developer (SRE)

- Engineered production **data and anomaly-detection pipelines** for IBM MQ transaction flows using **z-score analysis**, Prometheus, and Grafana, reducing root-cause analysis time by **40%** and preventing **10+ potential failures per month**.
- Designed a centralized **Definition of Done dashboard** tracking **100,000+ daily EOD transactions** across **15+ financial mandates**, improving operational visibility and decision-making speed by **40%**.
- Centralized application telemetry from **50+ Java microservices** using Prometheus Pushgateway and PromQL, enabling Grafana-based monitoring and faster incident diagnosis while reducing **mean-time-to-diagnose by 10%**.
- Partnered with **PM stakeholders** to develop an automated **alert-parsing pipeline**, integrating enterprise warehouse data with **regex-based asset detection** to generate and route **Jira tickets** to relevant squads, reducing operational TOIL by **30%**.
- Applied **Isolation Forest and Gaussian Mixture Models** to production log streams for anomaly detection, improving **incident detection accuracy by 25%** and reducing manual investigation effort.

Technical Skills

Programming: Python, SQL, R
Data Engineering: ETL/ELT, Data Pipelines, Apache Kafka, Apache Spark, PySpark, Apache Airflow, dbt, Data Modeling, Dimensional Modeling, Star Schema, OLTP/OLAP, Data Warehousing
Databases & Lakehouse: PostgreSQL, MySQL, Snowflake, Databricks, Delta Lake, Unity Catalog
Frameworks & Libraries: Pandas, NumPy, SQLAlchemy, FastAPI, Docker, Streamlit, Scikit-Learn, LLM, Flask, Matplotlib
Cloud & Tools: AWS, Git, Jenkins, Prometheus, Grafana, MLflow, GCP, Statistical Analysis, Render, Linux

Projects

E-Commerce Lakehouse Analytics Platform (2026) |

- Architected a production-scale **Databricks Lakehouse** processing **110M+ behavioral events** (**13.7GB**) from **5.3M users**, designing a **Bronze–Silver–Gold** architecture for scalable analytics and ML workloads.
- Engineered distributed **PySpark** ETL pipelines with **Delta Lake**, **Unity Catalog**, and **Spark SQL** for schema evolution, ACID transactions, partitioning, and incremental processing.
- Integrated **Apache Kafka** and **Spark Structured Streaming** to ingest and process continuous e-commerce events, extending the platform with near-real-time data processing alongside historical batch workloads.
- Created analytics-ready **dimensional datasets** through data validation, deduplication, sessionization, and business aggregations, supporting customer behavior, conversion funnels, revenue, and product analytics across **235K+ products**.

Supply Chain Data Pipeline (2026) |

- Architected an end-to-end **data engineering pipeline** ingesting **600,000+ supply chain records** into **PostgreSQL** using modular **Python**, **Pandas**, and **SQLAlchemy** ingestion workflows.
- Modeled a normalized **3NF OLTP database** from raw CSV data, enforcing relational integrity and eliminating redundancy through SQL-based schema design.
- Created a **dbt** transformation layer to convert normalized operational data into an analytics-ready **star schema**, producing reusable **fact** and **dimension** models for supply chain analytics.
- Integrated automated **dbt data quality tests**, version-controlled SQL transformations, and deployed an interactive **Streamlit** dashboard on **Render**.

HiveBlot – 2nd Place, UCSF Quantitative Biosciences Institute (QBI) Hackathon | Python | OCR | Computer Vision | LLMs

- Designed a curated **One Big Table (OBT)** data model by transforming OCR and LLM-extracted **Western blot experimental evidence** into a unified schema, enabling rapid evidence retrieval across **15+ biomedical research papers**.
- Orchestrated a modular extraction workflow integrating **OCR**, computer vision, and LLM-based information extraction to identify proteins, experimental conditions, biological outcomes, and associated figures across **15+ biomedical research papers**.
- Delivered a centralized evidence indexing system enabling researchers to retrieve experimental findings in seconds instead of manually reviewing **15+ papers** (saving an estimated **2–3 hours** per literature search), with traceability to original figures.

Online Retail Analysis (2025) |

- Performed feature engineering (RFM analysis, one-hot encoding) and data preprocessing on **500K+ transactions**, improving model performance and training efficiency.
- Evaluated **Logistic Regression**, **XGBoost**, and **Random Forest**; selected **Random Forest** based on highest **F1-score** and superior robustness to class imbalance.
- Deployed the best-performing model via **FastAPI** for real-time inference, containerized with **Docker** and hosted on **Render**, enabling scalable predictions for high-value customer targeting.

Education

University at Buffalo, SUNY *Aug 2025 – Dec 2026*
M.S. Data Science
Courses: Machine Learning, Statistical Learning, Probability, Data Models and Query Language, Reinforcement Learning

Vasantdada Patil Pratisthan's College of Engineering and Visual Arts *Jun 2017 – Jun 2021*
B.E. Information Technology